Digital Mohalla — deployment diagram, iteration 1, as built (v1.0, 14 October 2026; replaces the provisional Part A design A13)

UML 2.5 deployment notation: 3-D box = node («device» or «cloud service»); plain box = «executionEnvironment»; box with document icon = «artifact»; folded note = comment; solid line = communication path with protocol and port; dashed arrow = deployment trigger; grey dashed note = planned, not built.

«device»
Android phone [*]: residents, lenders, moderators

«executionEnvironment»
Expo runtime (Expo Go / development build)

«artifact» **digital-mohalla-app**
one app, screens by role (W1–W7); API address set at build time

«device»
Computer or phone with a web browser [*]

«executionEnvironment»
Web browser

«artifact» **web app**
the same Expo app, exported for the web and loaded from the static site

«cloud service»
GitHub

«artifact» **repository** Areeba444/DigitalMohalla
main = deployed, *develop* = integration; *render.yaml*

«executionEnvironment» **GitHub Actions runner**
«artifact» *ci.yml* on every push and pull request: Backend (ruff, mypy, bandit, pytest on a PostgreSQL 16 service container), Frontend (tsc, ESLint, Jest), API client in sync

HTTPS :443
REST /api/v1 (JSON)

HTTPS :443
REST /api/v1
(CORS)

HTTPS :443
static files

«cloud service»
Render

free plan, region Singapore;
HTTPS :443 ends at Render's edge (NFR-S-01)

«executionEnvironment»
Web service *digitalmohalla-api*: Docker container (sleeps after about 15 min idle)

«artifact» **API image** (*backend/Dockerfile*)
Python 3.12, FastAPI + Uvicorn, modular monolith (Modules 1, 3, 5, 9)
start-up: *alembic upgrade head*, then Uvicorn
REST */api/v1*; *GET /health*

«artifact» *config/engine.json*
config_version 1.3.0-iteration1-draft

Secrets in Render settings: *DATABASE_URL*, *JWT_SECRET* (generated); *CORS_ORIGINS* = the web app only

«executionEnvironment»
Static site *digitalmohalla-web* on Render's CDN (does not sleep)

«artifact» **Expo web export** (*frontend/dist*)
every path rewritten to *index.html*; API address built in

Render builds *main* only after the CI checks pass, and only when *backend/* or *frontend/* changed. A new API version goes live only if */health* answers; otherwise the previous one stays. Rollback: redeploy an earlier deploy (NFR-D-03, NFR-D-04).

push to main, CI passed: build and deploy

«device»
Team laptop: development and demo fallback

«executionEnvironment» **local run**
PostgreSQL 16, Uvicorn API, Expo dev server; synthetic seed data. Used if the hosted service is unavailable during the demo.

Planned, not built yet (Iteration 2): Android release build with the API address (EAS, TC-73); post-deploy smoke test (TC-91); rehearsed rollback; a paid tier or small host for the pilot if free-tier sleep affects users.

PostgreSQL protocol over TLS :5432
(sslmode=require)

«cloud service»
Neon

free plan, AWS ap-southeast-1 (Singapore), the same region as the API

«executionEnvironment» **PostgreSQL 16** with *btree_gist*; direct endpoint (not the pooler); suspends when idle, wakes on connect

«artifact» **schema *dm*** v2.3 (Alembic migration 0001): event store (append-only, hash-chained), projections, bookings (EXCLUDE), assessments, cases, moderator decisions, audit log. No coordinates, addresses or CNIC (suite 14).

Why: managed services give HTTPS, TLS to the database, health-gated deploys and rollback without server administration. Serves NFR-S-01, NFR-D-03, NFR-D-04, NFR-PR-01, NFR-S-07. The Part A design (one AWS host) is kept for reference.